

Attempt #3

Modifier Equation for the $P(n)$ Equation

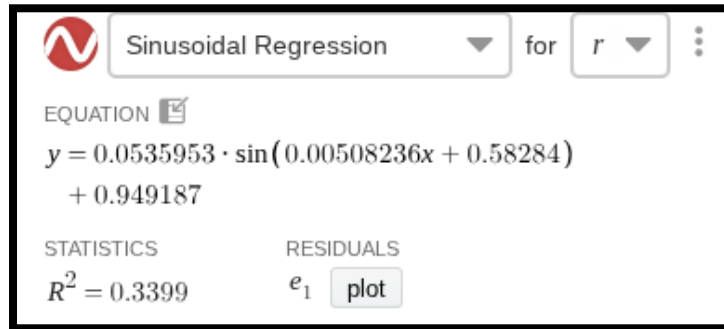
06/06/2026 - 10/06/2026

My third attempt expanded upon the $P(n)$ equation, shifting the focus away from the $S_p(n)$ formula and into the $P(n)$ equation. I postulated that, perchance, there was a multiplier equation that could multiply into the $P(n)$ equation to adjust for the differences between $P(n)$ and the actual prime at n .

By dividing the prime generated by $P(n)$ with the actual n^{th} prime integer (between $n = 10$ and $n = 2000$), I get the following table:

n	 $P(n)$	 A_p	 r
10	31	29	$\frac{29}{31}$
20	74	71	$\frac{71}{74}$
30	122	113	$\frac{113}{122}$
40	174	173	$\frac{173}{174}$
50	229	229	$\frac{229}{229}$
60	285	281	$\frac{281}{285}$
70	344	349	$\frac{349}{344}$
80	404	409	$\frac{409}{404}$
90	467	463	$\frac{463}{467}$
100	531	541	$\frac{541}{531}$

Generating a regression equation for the r column, I get the following:



Although the R^2 value is not exactly equal to 1, it's still a proof of concept. I then put the equation into our $P(n)$ equation in the following form:

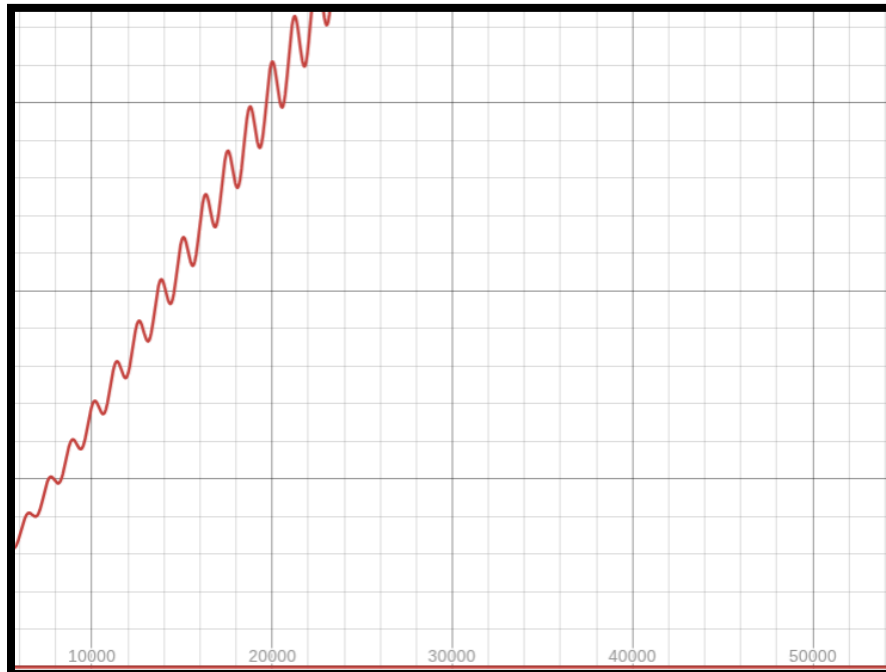
$$P(n) = \text{round} \left(f(n) \cdot \left(\frac{1}{S_p(n)} - \frac{1}{S_p(n-1)} \right) \right)$$

Where:

$$S_p(n) = \frac{1}{2 + \text{floor}(0.928914n^{2.20778})}$$

The rationale behind choosing the second $S_p(n)$ equation generated rather than the first was because the second $S_p(n)$ equation generated more accurate integers than the first $S_p(n)$ equation.

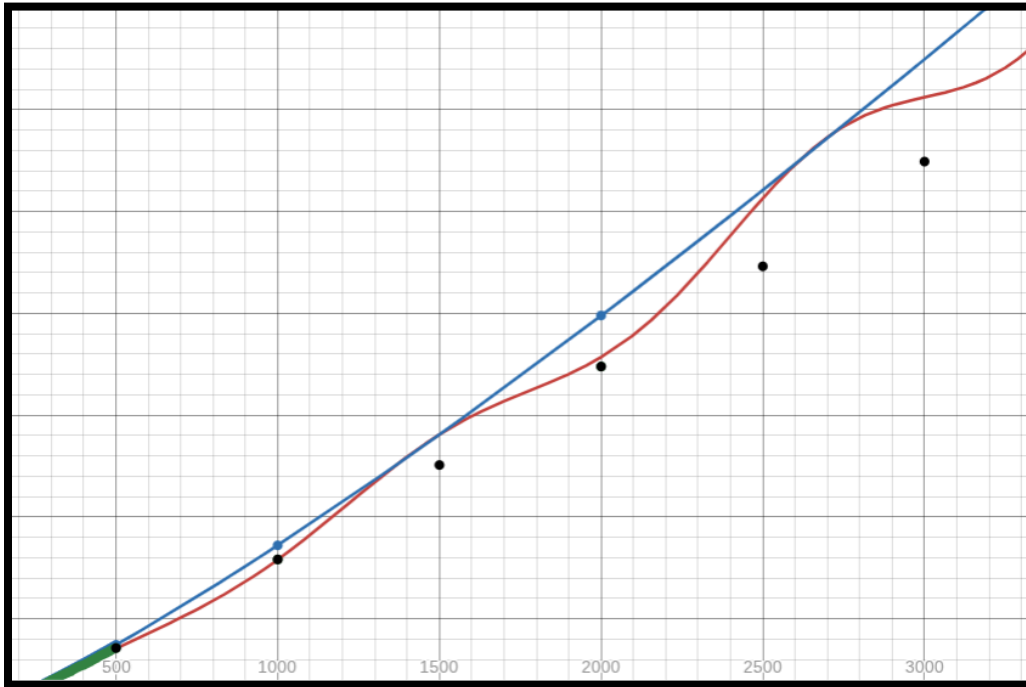
The following graph was generated using the sinusoidal modifier equation to the original $P(n)$ equation.



As seen below, the new $P(n)$ equation generated is more accurate than the previous $P(n)$ equation generated by Attempt #2.

n	 $P_{old}(n)$	 $P_{new}(n)$	 A_p
500	3725	3539	3571
1000	8610	7905	7919
1500	14054	14047	12553
2000	19894	17850	17389
2500	26048	25647	22307
3000	32467	30606	27449

Unfortunately, there is still a vast difference between the primes generated by the $P(n)$ equation and the actual prime integers.



A new approach should be taken with Attempt #4, straying away from the proposed approach assumed in Attempt #2. Although the sinusoidal modifier equation assisted in the correction of the $P(n)$ equation created in Attempt #2, it is clear from this attempt that the modifier was insufficient in correcting the primes generated by the $P(n)$ equation. Perchance the creation of the n^{th} prime formula should be reimagined entirely.